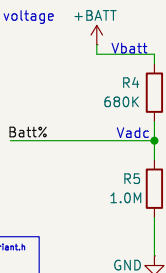


Battery voltage divider for measuring voltage

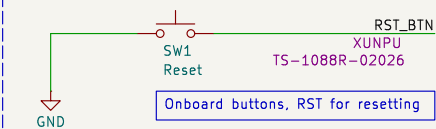
$R4 = 470$
 $R5 = 680$

Voltage Division Equation
 $V_{adc} = V_{batt} * (R5 / (R4 + R5))$
 # Voltage Vbatt --> Vadc ratio
 $ADC_Division = V_{adc} / V_{batt} = (R5 / (R4 + R5))$
 $R5 / (R4 + R5) = 0.591304347826087$
 # Voltage Vadc --> Vbatt ratio
 $ADC_Multiplier = V_{adc} / V_{batt} = (R4 + R5) / R5$
 $(R4 + R5) / R5 = 1.6913764705882353$

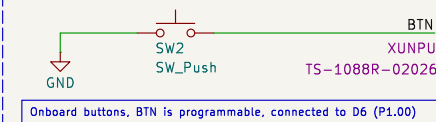


```
firmware/variants/nrf52840/dy/nrf52_promicro_diy_tcxo/variant.h
#define VBAT_DIVIDER_COMP (1.73)
#define ADC_MULTIPLIER VBAT_DIVIDER_COMP
```

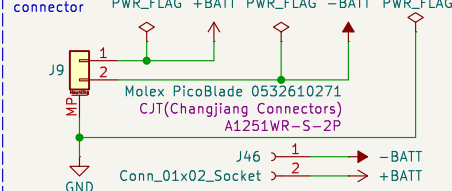
Onboard buttons RST



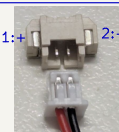
Onboard buttons BTN



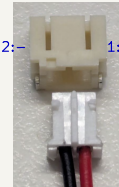
Battery connector



Molex PicoBlade 2-Pin
 - MPC: Molex, MPN: 0532610271
 - MPC: CJT (Changjiang Connectors), MPN: A1251WR-S-2P
 - Pitch: 1.25mm
 - Used By: HelTec LoRa 32
 - This polarity is also common for lipos in Aliexpress



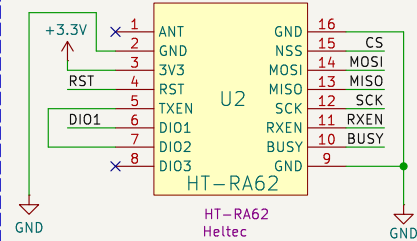
JST PH 2-Pin
 - MPN: S2B-PH-SM4-TB
 - Pitch: 2.00mm
 - Used By: RAK19007 WisBlock
 - This polarity is also common for lipos in Aliexpress



EXTRA GPIO



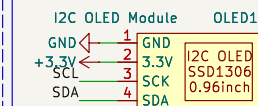
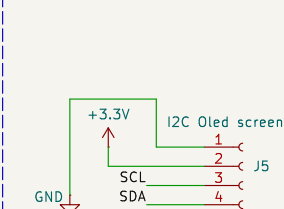
HT-RA62 / RA-015H



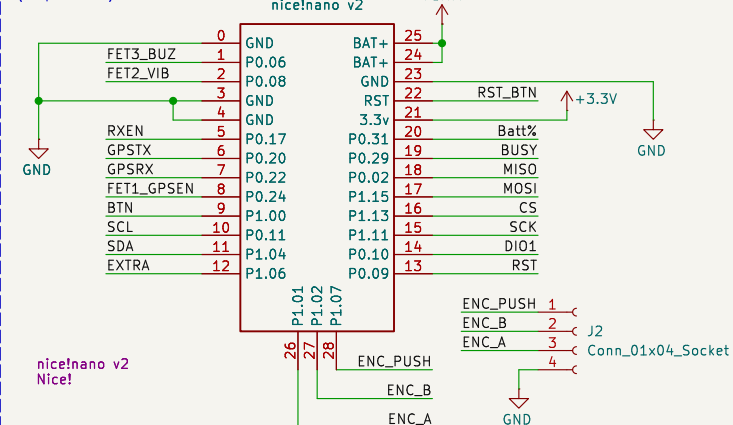
Note on DI02: DI02=1 during Tx and DI02=0 in any other mode, this is a special mode set during configuration. This is tied to TXEN to switch the internal RF TX switch. Switching the RXEN is done by the MCU to save the need for an extra inverter IC between DI02 and RXEN.

Note on DI01: DI01 is an interrupt that is triggered and outputs a rising edge when CadDetected and CadDone event is triggered. This basically means LORA channel activity was detected or finished.

Pins for I2C OLED screen

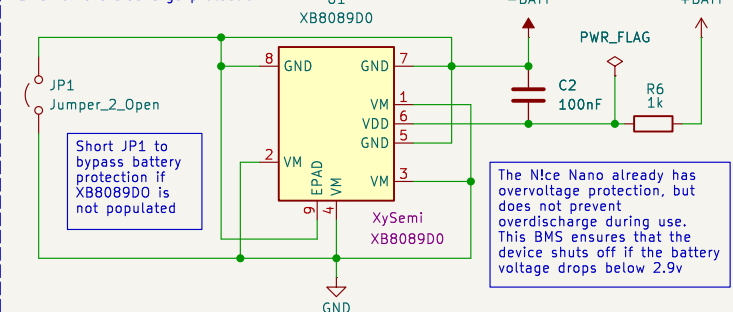


Nice!Nano nrf52840 microcontroller (or promicro)



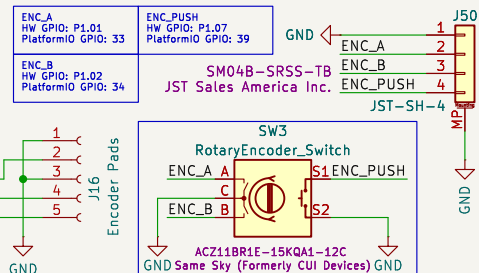
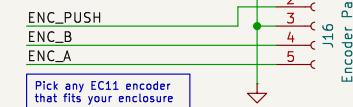
nice!nano v2
 Nice!

BMS for overdischarge protection

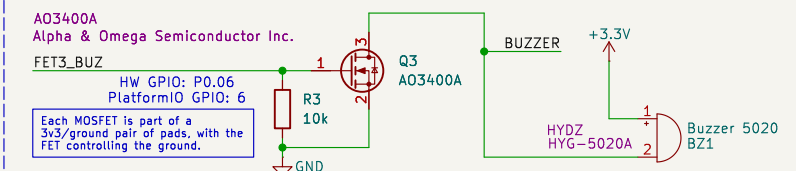
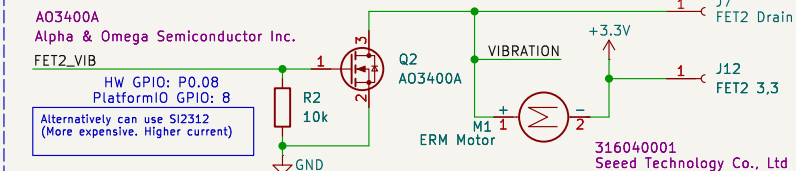
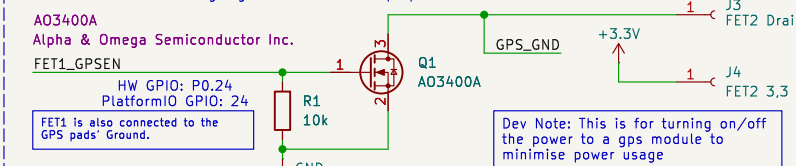


Pads for connecting encoder

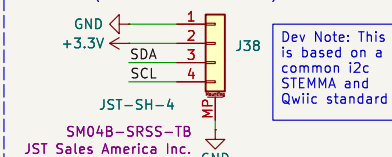
Dev Note: Pull ups not required as firmware appears to enable it internally. May be worth adding capacitor to A/B for debounce however.



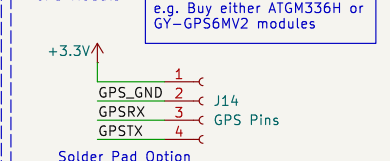
Mosfets used for switching high current external peripheral



I2C Port (For extra sensors etc...)



GPS Module



Development Source: https://github.com/mofosyne/roTec_pcb
 Fork Of FakeTec: https://github.com/jtljac/fakeTec_pcb
 Fork Of FakeTec: https://github.com/gargomoma/fakeTec_pcb/

Sheet: /
 File: fakeTecv5.kicad_sch

Title: roTecTH Rev5.4

Size: A4
 KiCad E.D.A. 9.0.6

Rev: 5.4
 Id: 1/1